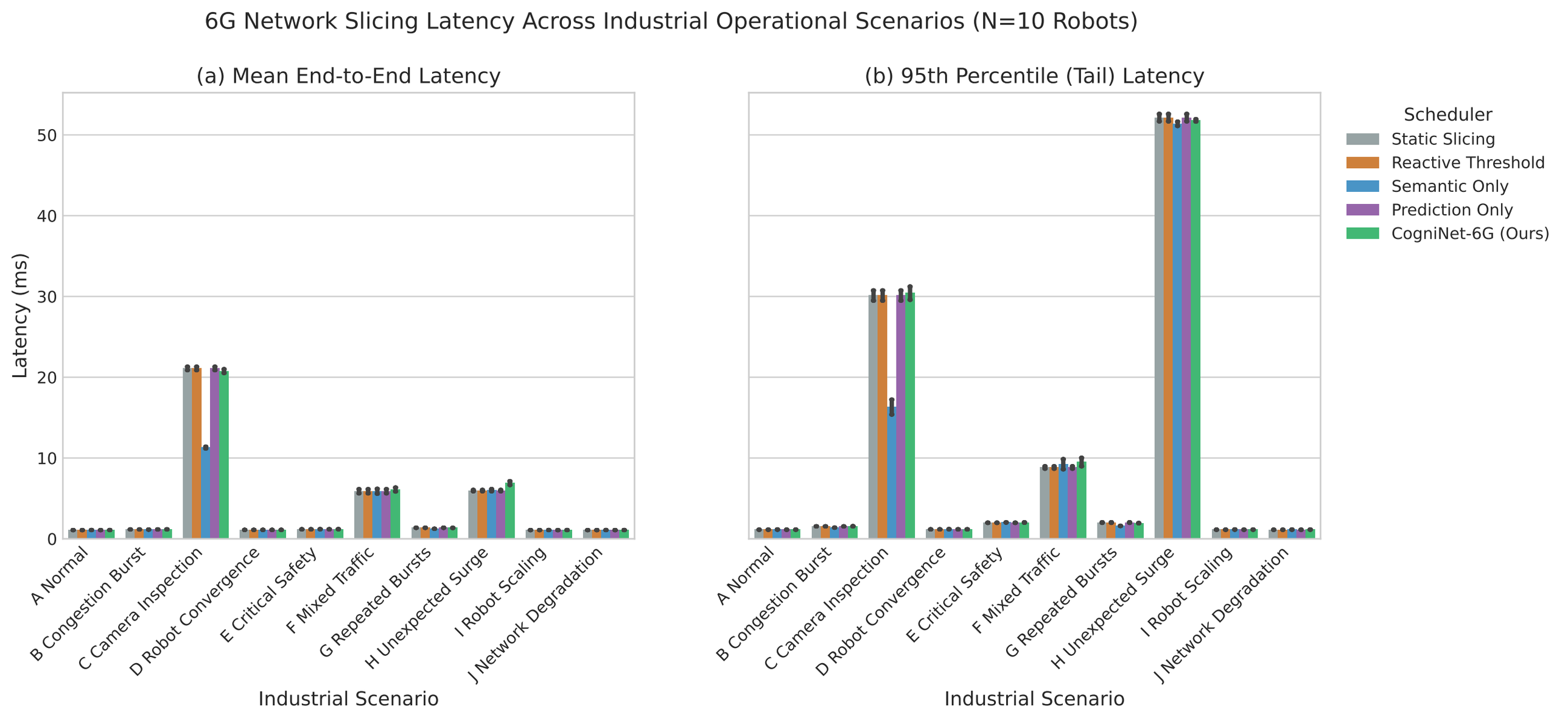
6G Network Slicing Latency Across Industrial Operational Scenarios (N=10 Robots)
(a) Mean End-to-End Latency
(b) 95th Percentile (Tail) Latency
Latency (ms)
0
10
20
30
40
50
A Normal
B Congestion Burst
C Camera Inspection
D Robot Convergence
E Critical Safety
F Mixed Traffic
G Repeated Bursts
H Unexpected Surge
I Robot Scaling
J Network Degradation
Industrial Scenario
Scheduler
Static Slicing
Reactive Threshold
Semantic Only
Prediction Only
CogniNet-6G (Ours)